

Java Program: FileNotFoundException Handling

1. Aim

To write a Java program that demonstrates a FileNotFoundException when a program attempts to open a file that does not exist and handles the exception using a try-catch block.

2. Description

A FileNotFoundException occurs when a program tries to access a file that cannot be found or opened. In this program, a File object is created for **missing_file.txt**. The FileReader attempts to open the file. Since the file is assumed to be unavailable, Java throws a FileNotFoundException. The catch block catches the exception and displays an appropriate error message.

3. Algorithm

1. Start the program.
2. Enter the main method.
3. Create a File object for the file named missing_file.txt.
4. Create a FileReader to open the file.
5. If the file cannot be found or opened, a FileNotFoundException is generated.
6. Catch the exception using a catch block.
7. Display a message indicating that the file could not be located.
8. Stop the program.

4. UML Diagram



5. Source Code with Comments

```
import java.io.File;
import java.io.FileReader;
import java.io.FileNotFoundException;

public class FileNotFoundException {
    public static void main(String[] args) {
        try {
            // Create a File object for a file that does not exist
            File file = new File("missing_file.txt");

            // Try to open the file for reading
            // This throws FileNotFoundException if the file is unavailable
            FileReader fr = new FileReader(file);

        } catch (FileNotFoundException e) {
            // Handle the file-not-found exception
            System.err.println(
                "Caught FileNotFoundException: "
                + "File could not be located."
            );
        }
    }
}
```

6. Compilation and Execution Commands

```
javac FileNotFoundException.java
java FileNotFoundException
```

The first command compiles the Java source file. The second command executes the compiled program.

7. Output Screenshot

FileNotFoundException - Console Output

```
Caught FileNotFoundException: File could not be located.
```

8. Result

The Java program was successfully executed. It demonstrated a `FileNotFoundException` when the program attempted to open the unavailable file `missing_file.txt`, and the exception was successfully handled using a try-catch block.